LightClearNet: Lightweight Encoder-Decoder Architecture for Single Image Dehazing

Kamaljot Singh
Department of Information Technology
Maharaja Agrasen Institute of Technology
New Delhi, India
kamaljotsingh.pgr@gmail.com

Md. Shadab Hussain
Department of Information Technology
Maharaja Agrasen Institute of Technology
New Delhi, India
shadabmyst@gmail.com

Seema Kalonia
Department of Information Technology
Maharaja Agrasen Institute of Technology
New Delhi, India
seema.kalonia25@gmail.com

Vandana Choudhary
Department of Information Technology
Maharaja Agrasen Institute of Technology
New Delhi, India
vandanachoudhari@gmail.com

Sunil Maggu
Department of Information Technology
Maharaja Agrasen Institute of Technology
New Delhi, India
sunilmaggu@mait.ac.in

Ajay Kumar Kaushik
Department of Information Technology
Maharaja Agrasen Institute of Technology
New Delhi, India
ajayk08@gmail.com

***Abstract*—Image dehazing is a crucial low-level vision task that recovers clear images from hazy ones. In recent years, immense progress has been made in recovering the original image as accurately as possible, without any consideration for efficiency and performance on low-end hardware, making it a mere relic for image beautification. Image dehazing could be very critical for enhanced visibility during indoor emergencies like fire hazards, utilization in surveillance and security cameras and in autonomous vehicles, and hence should be quick and easy to compute in order to be accessible. Therefore, we propose LightClearNet, a lightweight yet just as effective model for single image dehazing based around batch normalization and skip connections. The model has been trained on RESIDE-6K dataset and our lightweight model delivers comparable results to DehazeFormer while being twice as performant as the lightest variant of DehazeFormer.**

Keywords—image dehazing, real-time video dehazing, haze removal, visibility enhancement, computer vision

I. INTRODUCTION

Image dehazing is an integral computer vision task that focuses on enhancing the visual quality of images degraded by atmospheric haze, fog or smog. Ordinary dehazing methodologies regularly depend on complex models, making them computationally inefficient and challenging to pass on in real-world applications. The advancement of machine learning has enabled more capable and performance-oriented single-image dehazing solutions.

In this paper, we propose LightClearNet, a lightweight encoder-decoder designed for single-image dehazing. Unlike past approaches that frequently utilize complex structures, LightClearNet is made to be computationally viable while offering exceptional dehazing performance. Our approach leverages encoder-decoder architecture to effectively restore visibility and enhance image quality. Encoder-decoder architectures, inspired by the U-Net [1] architecture, consist of an encoder network that extracts high-level features from the input image and a decoder network that generates the dehazed output image based on the extracted features. This architecture facilitates the learning of complex mappings between hazy and haze-free images, enabling the model to effectively capture and remove the effects of atmospheric haze. We demonstrate the effectiveness of our method through extensive experiments on the RESIDE-6K dataset and real-world hazy images, showcasing its superior performance compared to existing dehazing techniques.

II. RELATED WORKS

The atmospheric scattering model [2]–[4] provides insights into the formation of hazy images. Formally, this model can be expressed as:

$$I(x) = J(x)t(x) + \alpha\big(1 - t(x)\big), \tag{1}$$

where $I(x)$ denotes the image obscured by haze, $J(x)$ represents the ground truth to be recovered, $t(x)$ depicts the medium transmission map, α indicates the global atmospheric light, and x indexes the pixels in the observed hazy image. $J(x)$ is recoverable upon estimating α and $t(x)$.

The medium transmission map, denoted as $t(x)$, represents the segment of light that remains undispersed and reaches the lens. It is defined by the equation:

$$t(x) = e^{-\beta d(x)}, \tag{2}$$

where $d(x)$ represents the distance between the scene point and the lens, and β signifies the coefficient for scattering in the atmosphere.

DehazeNet [5] was the pioneering work that introduced the use of a CNN-based model for single-image dehazing, demonstrating superior performance compared to traditional methods. Subsequent studies have explored various CNN architectures, including DenseNet [6] and U-Net [1], for image dehazing. These models leverage datasets like RESIDE [7] tailored to the dehazing task to learn effective representations of hazy scenes.

Our major inspiration for LightClearNet have been two state-of-the-art models for image dehazing, namely DehazeFormer [8] and MixDehazeNet [9]. DehazeFormer [8] utilizes a vision transformer technique that has achieved breakthroughs in various high-level vision tasks. This new method builds upon a popular vision transformer architecture called Swin Transformer [10]. However, it incorporates specific modifications to the network, including adjustments to the activation function (SoftReLU [8]), normalization layer, and how spatial information is processed. These changes make DehazeFormer [8] particularly adept at handling haze removal. DehazeFormer [8] has been shown to outperform

979-83503-6684-6/24/$31.00 ©2024 IEEE

existing methods on standard benchmarks, but the performance overhead makes it far from usable.

Similarly, MixDehazeNet [9] is also heavily inspired by DehazeFormer [8], but employs an encoder-decoder architecture, similar to ours. MixDehazeNet [9] also incorporates an innovative mixed attention mechanism, which operates at both spatial and channel levels. This mechanism enables the model to dynamically select and fuse features from different scales and channels, allowing it to focus on relevant information while suppressing irrelevant or noisy signals. MixDehazeNet [9] also suffers the same shortcomings of suboptimal performance, making real-time dehazing a distant dream.

III. Methodology

Hazy image (input) Clear image (output)

Fig. 1. RESIDE-6K

The dataset used for this research is RESIDE-6K. Samples from the dataset are shown in Fig. 1. RESIDE-6K is a subset of the RESIDE (REalistic Single Image DEhazing) [7] dataset. Each image pair in RESIDE-6K includes a hazy image and its corresponding haze-free ground truth. The dataset consists of 6,000 image pairs in the training set and 1,000 image pairs in the testing set. The training dataset consists of 3,000 pairs of indoor images (ITS) and 3,000 pairs of outdoor images (OTS), all resized to 400×400 pixels. Similarly, the testing set also combines indoor and outdoor image pairs, forming a test set of 1,000 image pairs, with 500 pairs each for indoor and outdoor scenes, without any resizing. This split has been demonstrated in Table I. Due to not resizing the images in the test set, the images can't be stacked up due varying inner dimensions, leading us to setting the batch size for test data-loader equal to 1.

The training set is further split into two sets to accommodate for the validation set, with 500 image pairs randomly selected for validation out of the initial 6,000 image pairs. So, the final split size of the overall dataset is $(5500, 500, 1000)$ for training, validation and testing sets respectively.

TABLE I. RESIDE-6K Breakdown

Splits	No. of image pairs		
	Total	***Indoor (ITS)***	***Outdoor (OTS)***
Train	6,000	3,000	3,000
Test	1,000	500	500

Our goal with LightClearNet was to develop a lightweight encoder-decoder architecture for single-image dehazing capable of performing dehazing even on underpowered hardware. The overall architecture of LightClearNet has been shown in Fig. 2.

The model has been trained on a modestly equipped PC as well as on Google Colab. PC hardware and software configuration:

a) Hardware

- **CPU:** Ryzen 5 5500
- **GPU:** Nvidia Geforce 1660 Ti
- **RAM:** 32GB DDR4 3200MHz

b) Software

- **OS:** Windows 11 23H2
- CUDA 12.4
- Python 3.11.7
- PyTorch 2.2.0
- Torchvision 0.17.0
- TorchMetrics 1.3.1
- OpenCV 4.9.0

The model has been trained with a batch size of 4 for train as well as validation data-loaders while on PC, and a batch size of 16 while on Google Colab, taking advantage of the extra VRAM Colab's T4 provides (1660 Ti: 6GB, T4: 15GB). We have made sure to write not only device-agnostic, but also platform-agnostic code as best as we could, to easily switch between local machines and Colab.

The core of the model is centered around convolutions, which are pivotal for pattern recognition in images. Convolutions serve as the foundation in both image processing and deep learning. They comprise of the application of a small matrix, referred to as a kernel or filter, onto an input image. This filter is a small matrix of numbers that slides (or convolves) over the input image pixel by pixel, performing a dot product at each position. The result of this dot product is then placed into the corresponding position in the output image, forming a new image known as the feature map or convolutional feature. Secondly, Batch Normalization (BatchNorm) [11] plays a crucial role in enhancing the model's learning capabilities and minimizing the loss. BatchNorm [11] is a data preprocessing technique for neural networks which normalizes the inputs for every layer by subtracting their mean and dividing by their standard deviation within a mini-batch. This normalization step helps keep the inputs to each layer within a reasonable range, which prevents the network's activations from becoming too large or too small. Furthermore, skip connections, inspired by ResNet [12], enable the model to better preserve and recover fine

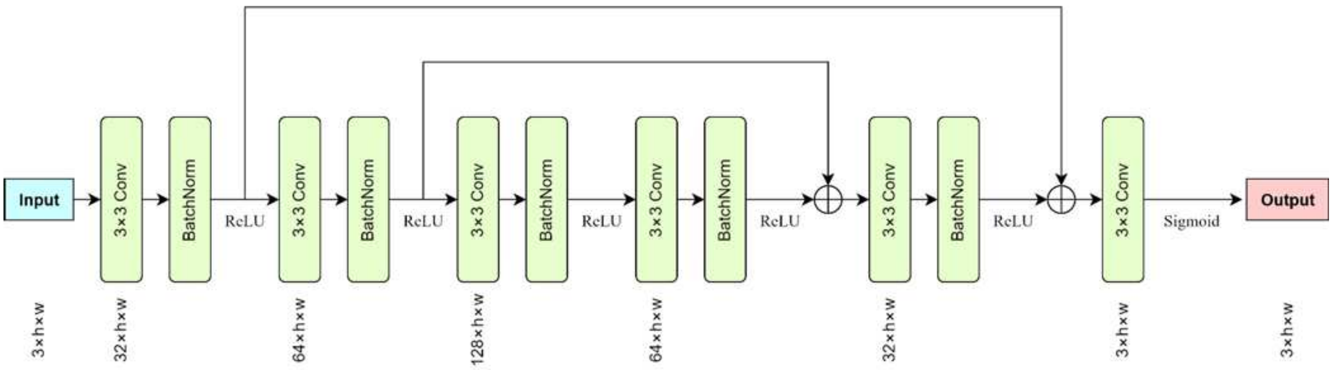


Fig. 2. LightClearNet Architecture

details and structural information in the image, improving the SSIM. At the end, the output is normalized via ***sigmoid*** activation function in order to get all the values within the range 0 to 1, as PyTorch expects image tensors to be of $dtype = torch.float32$, in contrast to NumPy which expects $uint8$.

We've prioritized maintaining image resolution throughout our process, avoiding downsampling to prevent significant deterioration in image quality. Our architecture doesn't require image cropping either, as it doesn't flatten internal neural layers, enabling our model to handle images of any dimensions effectively.

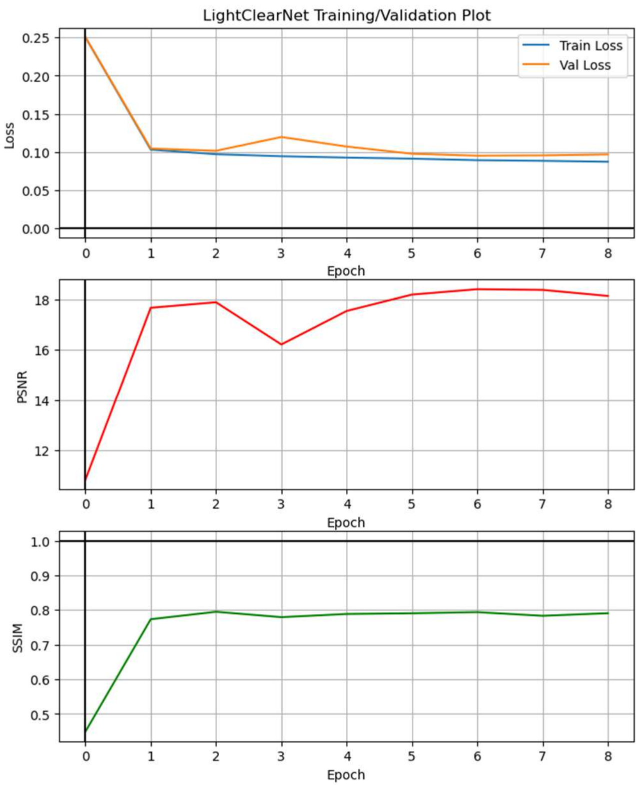


Fig. 3. LightClearNet Training/Validation Plot

Due to the findings in **Error! Reference source not found.**, the criterion used is L1 loss (MAE), as it outperforms L2 loss (MSE) in image-based regression tasks, and the optimizer is ADAM configured with the default values of 0.9 and 0.999 for β_1 and β_2, and the learning rate set to the default value of 0.001. Experiments have been done with a learning rate of 10^{-4} as well as 10^{-6}, without any improvement in the quality of the output, while being slower to train. The utilities for calculating PSNR and SSIM come from TorchMetrics's *"image"* subpackage.

The model has been extensively trained multiple times up to 70 epochs. However, the model is very prone to overtraining. The metrics don't improve at all, if not worsen and the image quality of the model's output deteriorates significantly, with unpleasant to look at patches on flat surfaces. We found training around 5-7 epochs to be the sweet spot. This training/validation process has been shown in Fig. 3.

IV. Experimental Results

A. *Quantitative Comparison and Run-time Analysis*

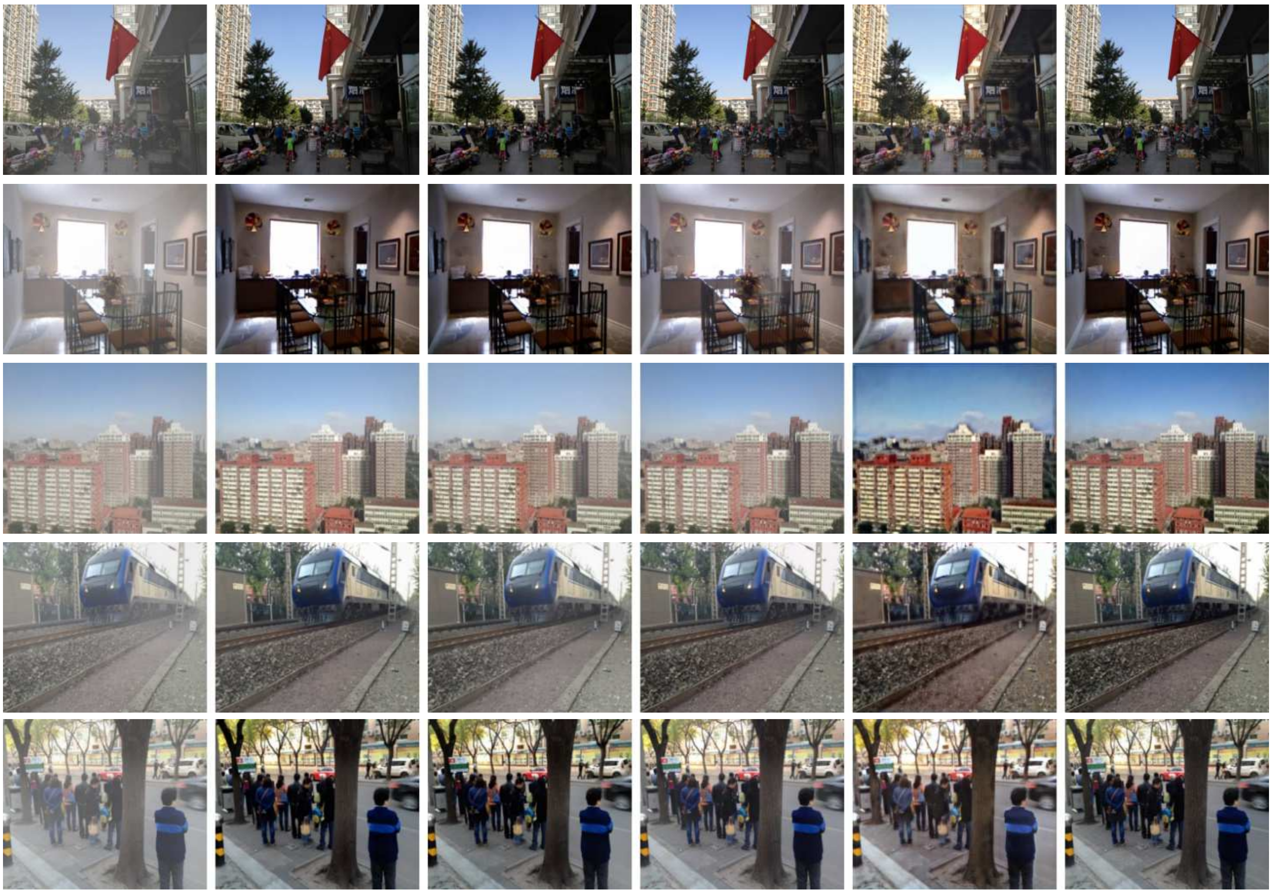
All our testing has been done on the test set of RESIDE-6K. We independently tested the different variants of DehazeFormer [8] and MixDehazeNet [9] as well to determine PSNR and SSIM. The pretrained weights for these models have been shared by their original creators via Google Drive for open use. The *"reside6k"* weights variant has been used for DehazeFormer [8] and the *"outdoor"* variant has been used for MixDehazeNet [9]. All the models are rigorously checked for performance on the local system for real-time dehazing using OpenCV. Frames-per-second (FPS) have been determined, while running the models on the local system for around 2 minutes, with higher numbers indicating better performance. With higher FPS, the transition between consecutive frames appears seamless, making the resulting video feel more fluid and natural. These results are shown in Table II, with **bold** for the best results and underline for the second-best in every metric.

It is clear from these results that our model offers almost twice the performance as compared to the lightest variant among the other models (DehazeFormer-T [8]), and a whopping 6.2 times performance increase over the best quality model (DehazeFormer-B [8]). Our model ends up showing comparable image quality with little to no difference in achieving the end goal of dehazing, which is enhancing the visibility in hazy environments, especially during emergencies.

Also we figured out that during real-time dehazing, our model was getting CPU-bottlenecked, in contrast to other models which were experiencing a GPU-bottleneck, meaning our model's dependence on strong graphical processing power is relatively minimal, and the FPS can increase further if given a capable multi-core CPU. And the difference in performance only increases in the absence of a dedicated graphics

TABLE II. QUANTITATIVE COMPARISON OF VARIOUS DEHAZING METHODS ON THE RESIDE-6K DATASET

Methods	**RESIDE-6K**		
	PSNR	***SSIM***	***FPS***
DehazeFormer-B [8]	**25.88**	**0.940**	1.91
DehazeFormer-M [8]	25.64	0.936	2.46
DehazeFormer-S [8]	<u>25.80</u>	<u>0.939</u>	3.90
DehazeFormer-T [8]	24.69	0.929	<u>6.85</u>
MixDehazeNet-L [9]	25.58	0.922	1.05
MixDehazeNet-B [9]	25.75	0.928	1.98
MixDehazeNet-S [9]	22.75	0.892	3.53
(ours) LightClearNet	21.17	0.852	**11.94**



Input DehazeFormer-B DehazeFormer-S MixDehazeNet-L LightClearNet GT

Fig. 4. Qualitative comparison of various dehazing methods on the RESIDE-6K dataset

processor. This is a major reason why our model is much more accessible as compared to other models, as it can perform great even on budget-friendly PCs and laptops which often lack dedicated GPUs, making it ideal for a wider range of users. Moreover, with sufficient optimization, it has the potential to run smoothly on mobile devices, further extending its reach to users on the go. This versatility ensures that our model can be deployed across various platforms without compromising its performance.

When considering storage requirements, the state-dict of our model is impressively compact at only 745 KB, whereas alternative models like MixDehazeNet-L [9] can reach up to 143 MB in size, with the best quality model DehazeFormer-B [8] still weighing around 15 times as much at 10.7 MB, further accentuating the point that our model achieves comparable results while maintaining an exponentially smaller size and requiring drastically less computational power.

B. Qualitative Comparison

Every model's results look remarkably good, with all of them achieving the primary objective of enhancing visibility and improving the image quality by removing the effects of atmospheric haze while minimizing artifacts and distortions. This qualitative comparison has been shown in Fig. 4. Color Fidelity is also exceptionally great, with all the models preserving the original colors of the scene as much as possible. Our model specifically beats others in contrast enhancement, which ends up improving visibility and perceptual clarity of the scene. Effective contrast enhancement makes it easier to discern details and structures in hazy images.

V. Conclusion

This paper proposes LightClearNet, a lightweight yet highly effective model for single-image dehazing. While existing image dehazing methodologies often prioritize quality over efficiency, LightClearNet is designed to be computationally viable without compromising on performance. Leveraging an encoder-decoder architecture, LightClearNet effectively restores visibility and enhances image quality by extracting high-level features from the input image and generating the dehazed output image based on these features. Our experiments on the RESIDE-6K dataset and real-world hazy images demonstrate LightClearNet's superior performance, demonstrating comparable results to more complex models like DehazeFormer [8] while delivering twice the performance as the lightest variant among the other models, with an impressive PSNR of 21.17 dB and SSIM of 85.24%. These findings underscore LightClearNet's potential for practical deployment in various real-world scenarios, addressing the critical need for quick and accessible image dehazing solutions beyond just image beautification.

References

[1] Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." In Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18, pp. 234-241. Springer International Publishing, 2015.

[2] McCartney, Earl J. "Optics of the atmosphere: scattering by molecules and particles." New York (1976).

[3] Nayar, Shree K., and Srinivasa G. Narasimhan. "Vision in bad weather." In Proceedings of the seventh IEEE international conference on computer vision, vol. 2, pp. 820-827. IEEE, 1999.

[4] Narasimhan, Srinivasa G., and Shree K. Nayar. "Vision and the atmosphere." International journal of computer vision 48 (2002): 233-254.

[5] Cai, Bolun, Xiangmin Xu, Kui Jia, Chunmei Qing, and Dacheng Tao. "Dehazenet: An end-to-end system for single image haze removal." IEEE transactions on image processing 25, no. 11 (2016): 5187-5198.

[6] Huang, Gao, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q. Weinberger. "Densely connected convolutional networks." In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 4700-4708. 2017.

[7] Li, Boyi, Wenqi Ren, Dengpan Fu, Dacheng Tao, Dan Feng, Wenjun Zeng, and Zhangyang Wang. "Benchmarking single-image dehazing and beyond." IEEE Transactions on Image Processing 28, no. 1 (2018): 492-505.

[8] Song, Yuda, Zhuqing He, Hui Qian, and Xin Du. "Vision transformers for single image dehazing." IEEE Transactions on Image Processing 32 (2023): 1927-1941.

[9] Lu, LiPing, Qian Xiong, DuanFeng Chu, and BingRong Xu. "MixDehazeNet: Mix Structure Block For Image Dehazing Network." arXiv preprint arXiv:2305.17654 (2023).

[10] Liu, Ze, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. "Swin transformer: Hierarchical vision transformer using shifted windows." In Proceedings of the IEEE/CVF international conference on computer vision, pp. 10012-10022. 2021.

[11] Ioffe, Sergey, and Christian Szegedy. "Batch normalization: Accelerating deep network training by reducing internal covariate shift." In *International conference on machine learning*, pp. 448-456. pmlr, 2015.

[12] He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. "Deep residual learning for image recognition." In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 770-778. 2016.

[13] Lim, Bee, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. "Enhanced deep residual networks for single image super-resolution." In Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp. 136-144. 2017.